

Digifort RTSP Interface

For Digifort 7.3.0

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1 Overview

1.1 Description



This document specifies the Digifort media interface based on RTSP protocol.

This interface provides access to live video from cameras of a Digifort server using the RTSP protocol, allowing for any media player that supports this protocol to receive live video from Digifort servers. This interface can also be used as a tool for systems integration.

1.2 History

Digifort Version	Comments
6.7.0.0	First version
7.2.1.0	Added playback support
7.3.0.0	- Added support for RTP Header Extension with absolute timestamp information - Added <code>SourceStorage=Edge</code> parameter for Edge Playback

2 Definitions

This section contains general information related to the use of this document

2.1 General notations

2.1.1 General abbreviations

The following abbreviations are used in this document

Abbreviation	Description
N/A	Not applicable - The feature/parameter/value is not used in the specified task
URL	Uniform Resources Location (URL) is a compact string that represents a feature available in Internet. The RFC 1738 describes the syntax and semantics for a URL
URI	Uniform Resource Identifier (URI) is a compact string of characters for identifying an abstract or physical feature. The RFC 3986 describes the generic syntax of a URI

2.1.2 Style conventions and general syntax of the URL

In URL syntax and in the descriptions of the arguments, text in italics with greater than and less than signs denote a content that must be substituted by a value or a string. When substituting the text, the greater than and less than signs must also be substituted. For example, the name of a camera is denoted by *<cameraname>* in description of URL syntax. In the example of URL syntax, *<cameraname>* is substituted by the string mycamera.

The URL syntax is written with the word "**Syntax:**" in bold face, accompanied by a box with the syntax of the command.

Syntax:

```
rtsp://<server_address>/Interface/<object>[/<subobject>...]/<command>
[?<argument=value>[&<argument=value>...]]
```

Example 1: Request live video from camera 1

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=1
```

2.2 Types of variables

This programming interface implements some formats of proper variables. The following are the variables and proper formats:

Variable	Type of data	Description
APIDate	Date values	This format of variable is applied where data values are passed directly in the URL. The format of date must be YYYY.MM.DD Ex: 2009, November, 15 2009.11.15
APITime	Time values	This format of variable is applied where time values are passed directly in the URL. The format of time must be HH.MM.SS Ex: 15:30:45 15.30.45

3 RTSP server implementation

This session provides details on the implementation of the Digifort RTSP server.

3.1 RFCs

The Digifort RTSP server was developed using as a basis the following RFCs:

Server base:

[RFC 2326](#) - Real Time Streaming Protocol (RTSP)

[RFC 2327](#) - Session Description Protocol (SDP)

Packetizers:

[RFC 3550](#) - RTP: A Transport Protocol for Real-Time Applications

[RFC 2435](#) - RTP Payload Format for JPEG-compressed Video

[RFC 3016](#) - RTP Payload Format for MPEG-4 Audio/Visual Streams

[RFC 3984](#) - RTP Payload Format for H.264 Video

[RFC 3551](#) - RTP Profile for Audio and Video Conferences with Minimal Control

[RFC 3640](#) - RTP Payload Format for Transport of MPEG-4 Elementary Streams

[RFC 7798](#) - RTP Payload Format for High Efficiency Video Coding (HEVC)

3.2 Supported RTSP commands

The Digifort RTSP server supports the following protocol commands:

OPTIONS

DESCRIBE

SETUP

PLAY

PAUSE

TEARDOWN

3.3 Supported media types

The Digifort RTSP server supports the packing and transmission of the following media types:

Video: Motion JPEG, MPEG-4, H.264, H.265

Audio: Linear PCM, G.711, G.726, AAC

3.4 Transmission protocol

The Digifort RTSP server allows the transmission of media using UNICAST over TCP or UDP protocols.

MULTICAST transmission is not yet supported.

3.5 Media transcoding

The Digifort RTSP server **DOES NOT** transcode the media, it only serves as a bridge to the original media stream sent by the camera, with it, you cannot request a media stream with resolution, frame rate or quality different from the values configured on the camera.

3.6 RTP Extension Headers

The RTSP server implements some custom RTP Extension Headers that can be included in the streaming (Live and Playback) to provide extra information on the stream. By default, no RTP Extension Header will be added to the stream, unless specified by `RTPExtHdr` parameter in the streaming URL

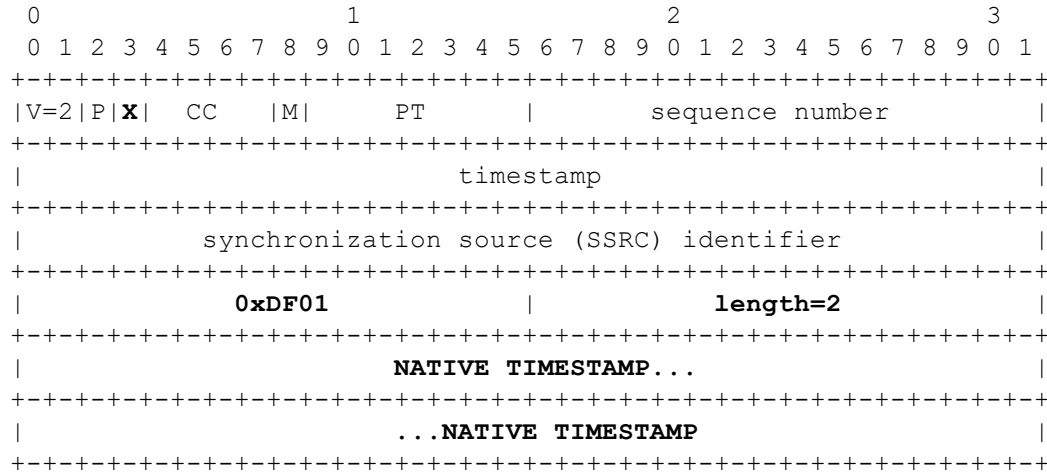
3.6.1 Native Timestamp

In order to allow clients to report a stable and accurate timestamp for each frame played back, it is necessary to associate an absolute timestamp with each packet, or each group of packets with the same RTP timestamp (e.g. a video frame).

This is achieved using an RTP header extension containing a Native Timestamp (Timestamp format used internally in the server).

The timestamp mechanism uses the extension ID 0xDF01 for the RTP extension.

Below shows the general form of an RTP packet containing this extension:



The fields of this extension are as follows:

- Native Timestamp. Is a type that maps to a 64 bit Double value. The integral part of a Native Timestamp value is the number of days that have passed since 12/30/1899. The fractional part of the Native Timestamp value is fraction of a 24 hour day that has elapsed. The Timestamp value is UTC. Although values in an RTP packet are usually transmitted in big-endian format, this value alone is transferred in **little-endian** format. The underlying Double type used to store the Native Timestamp is an IEEE standard floating-point type that uses 8 bytes to store a sign bit, an 11-bit exponent, and a 52-bit mantissa. The mantissa is usually normalized, that is, it has an implicit 1 bit before the most significant bit. If the exponent is zero, however, the mantissa is denormalized without the implicit 1 bit. Thus, the numerical value of +0.0 is represented by all zero bits. An exponent of all 1 bits represents infinity (mantissa is zero) or not-a-number (mantissa is not zero). The limits of the Double type are approximately 2.23×10^{-308} to 1.79×10^{308} , with about 15 decimal digits of precision.
- Extension Header ID is 0xDF01
- Extension Header Length is 2 (Number of 32bits words in the header)
- X is signaled as 1 to indicate the presence of an extension header

3.6.2 ONVIF Replay

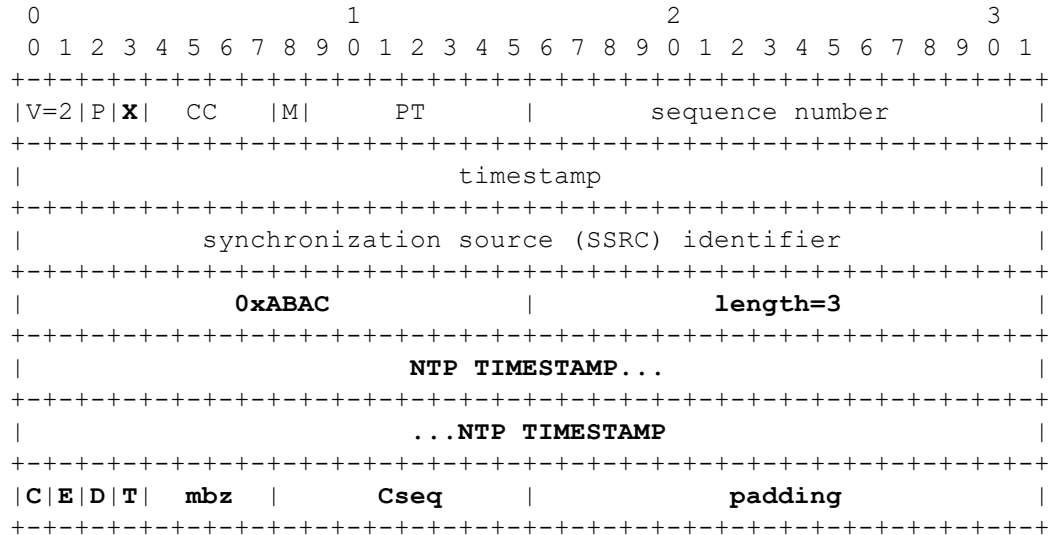
In order to allow clients to report a stable and accurate timestamp for each frame played back regardless of the direction of playback, it is necessary to associate an absolute timestamp with each packet, or each group of packets with the same RTP timestamp (e.g. a video frame).

This is achieved using an RTP header extension containing an NTP timestamp and some additional information also useful for replay.

This replay mechanism is defined by ONVIF Streaming Specification Version 19.06, June, 2019
<https://www.onvif.org/specs/stream/ONVIF-Streaming-Spec.pdf>

The replay mechanism uses the extension ID 0xABAC for the replay extension.

Below shows the general form of an RTP packet containing this extension:



used to initiate transmission. When a client sends multiple, consecutive PLAY commands, this value may be used to determine where the data from each new PLAY command begins. **The current implementation of our RTSP server does not implement this field (It will always send 0).**

4 Interface

This section contains information related to the API Interface

4.1 Requesting live video

The RTSP server from Digifort System allows for the request of live video from any camera of the server.

Security level: Requires user authentication through RTSP protocol

Syntax:

```
rtsp://<server_address>/Interface/Cameras/Media?<argument=value>
[&<argument=value>...]
```

Arguments:

Argument	Valid values	Description						
Camera*	String	Camera name						
Profile	Recording Visualization Custom	Type of media profile: Recording - Request the media stream using the default recording profile Visualization - Request the media stream using the default visualization profile Custom - Request the media stream using the profile specified on CustomProfile argument If this parameter is omitted, the default value Recording will be used						
CustomProfile	String	Media profile name in case the value of argument Profile is "Custom"						
RTPExtHdr	NativeTimestamp ONVIFReplay	Add an extension header to the RTP packet containing absolute timestamp information. If this parameter is omitted, no extension RTP header will be added. <table><tr><th>Name</th><th>Description</th></tr><tr><td>NativeTimestamp</td><td>Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.</td></tr><tr><td>ONVIFReplay</td><td>Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.</td></tr></table>	Name	Description	NativeTimestamp	Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.	ONVIFReplay	Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.
Name	Description							
NativeTimestamp	Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.							
ONVIFReplay	Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.							

* Mandatory parameters

Example 1: Request media stream from camera 1, using default recording profile

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=Camera1
```

or

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=Camera1&
Profile=Recording
```

Example 2: Request media stream from camera 2, using default visualization profile

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=Camera2&
Profile=Visualization
```

Example 3: Request media stream from camera ABC, using the previously registered and configured media profile, named "LowResolution"

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=ABC&
Profile=Custom&CustomProfile=LowResolution
```

Example 4: Request media stream from camera 2, using default visualization profile and ONVIF Replay RTP extension header

```
rtsp://192.168.0.1:554/Interface/Cameras/Media?Camera=Camera2&
Profile=Visualization&RTPExtHdr=ONVIFReplay
```

4.2 Requesting playback

The RTSP server from Digifort System allows for the request of playback from any camera of the server.

Security level: Requires user authentication through RTSP protocol

Syntax:

```
rtsp://<server_address>/Interface/Cameras/Playback?<argument=value>
[&<argument=value>...]
```

Arguments:

Argument	Valid values	Description
Camera*	String	Camera name
StartDate*	APIDate	Start date of recordings on APIDate format
StartTime*	APITime	Start time of recordings on APITime format
EndDate	APIDate	End date of recordings on APIDate format
		If this parameter is omitted, the current date will be used
EndTime	APITime	End time of recordings on APITime format
		If this parameter is omitted, the current time will be used
Push	TRUE FALSE	TRUE - Send recorded footage as fast as possible, with no playback speed control (Use when downloading recordings). FALSE - Send recorded footage with playback speed control of 1x If this parameter is omitted, the default value FALSE will be used
Audio	TRUE FALSE	TRUE - Send audio multiplexed with video (If audio is recorded) FALSE - Send video only If this parameter is omitted, the default value FALSE will be used
SourceStorage	HOT COLD EDGE	This parameter specifies the source storage used to retrieve the media.

		<table><tr><th>Name</th><th>Description</th></tr><tr><td>HOT</td><td>Hot storage is the main storage of the system, where all recordings are stored</td></tr><tr><td>COLD</td><td>Cold storage refers to archived recordings by the Archiving system</td></tr><tr><td>EDGE</td><td>Edge storage refers to the storage of recordings inside the device</td></tr></table>	Name	Description	HOT	Hot storage is the main storage of the system, where all recordings are stored	COLD	Cold storage refers to archived recordings by the Archiving system	EDGE	Edge storage refers to the storage of recordings inside the device
Name	Description									
HOT	Hot storage is the main storage of the system, where all recordings are stored									
COLD	Cold storage refers to archived recordings by the Archiving system									
EDGE	Edge storage refers to the storage of recordings inside the device									
		If this parameter is omitted, the default value HOT will be used.								
RTPExtHdr	NativeTimestamp ONVIFReplay	Add an extension header to the RTP packet containing absolute timestamp information. If this parameter is omitted, no extension RTP header will be added. <table><tr><th>Name</th><th>Description</th></tr><tr><td>NativeTimestamp</td><td>Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.</td></tr><tr><td>ONVIFReplay</td><td>Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.</td></tr></table>	Name	Description	NativeTimestamp	Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.	ONVIFReplay	Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.		
Name	Description									
NativeTimestamp	Add a NativeTimestamp extension header to RTP packets containing native absolute timestamp for each packet.									
ONVIFReplay	Add an ONVIF Replay extension header to RTP packets containing absolute timestamp for each packet.									

* Mandatory parameters

Example 1: Request recordings of camera "Camera1" from January 01, 2014 10:00:00AM to January 02, 2014 09:00:00AM with error response in XML

```
rtsp://192.168.0.1:554/Interface/Cameras/Playback?Camera=Camera1&
StartDate=2014.01.01&StartTime=10.00.00.000&EndDate=2014.01.02&
EndTime=09.00.00.000&ResponseFormat=XML
```

Example 2: Request recordings of camera "Camera1" from January 01, 2014 to date, with audio, without playback speed control and error response in text

```
rtsp://192.168.0.1:554/Interface/Cameras/Playback?Camera=Camera1&
StartDate=2014.01.01&StartTime=00.00.00.000&Audio=TRUE&Push=TRUE&
ResponseFormat=Text
```

Example 3: Request recordings of camera "Camera1" from January 25, 2014 to date, with audio and playback speed control

```
rtsp://192.168.0.1:554/Interface/Cameras/Playback?Camera=Camera1&
StartDate=2014.01.20&StartTime=00.00.00.000&Audio=TRUE
```

Example 4: Request recordings of camera "Camera1" from January 25, 2014 to date, with audio and playback speed control from cold storage

```
rtsp://192.168.0.1:554/Interface/Cameras/Playback?Camera=Camera1&
StartDate=2014.01.20&StartTime=00.00.00.000&Audio=TRUE&SourceStorage=COLD
```

Example 5: Request recordings of camera "Camera1" from January 01, 2014 to date, with audio and ONVIF Replay extension RTP Header

```
rtsp://192.168.0.1:554/Interface/Cameras/Playback?Camera=Camera1&
StartDate=2014.01.01&StartTime=00.00.00.000&Audio=TRUE&
RTPExtHdr=ONVIFReplay
```

5 Tested RTSP clients

The following RTSP clients were tested with Digifort RTSP server successfully:

VLC 2.0.6 - The worlds most popular free media player (www.videolan.org)

Wowza Media Server 3 - Powerful media server, that can be used for internet video broadcasting. This media server is capable of delivering video in different protocols, allowing for the use of different media players, including flash plugins that can be added to web pages (www.wowza.com)